

AARYAN PAWAR

MACHINE LEARNING • COMPUTER VISION • RESEARCH ENGINEERING

Bellevue, WA | LinkedIn | GitHub | Portfolio

SUMMARY

UW Computer Science student with hands-on experience in attention-based deep learning, computer vision, C++ performance engineering, and reproducible model evaluation. Builds practical systems across research, software, and embedded platforms; interested in applying transformer methods to computational biology and other ML-driven domains.

EDUCATION

University of Washington

B.S. Computer Science, Senior Standing

GPA 3.93/4.00 · Dean's List, six consecutive quarters

Seattle, WA

Expected June 2027

Incoming Undergraduate Teaching Assistant, CSE 421: Algorithms, Paul G. Allen School, UW Seattle

Autumn 2026

RESEARCH & ENGINEERING EXPERIENCE

Alitheon

Machine Learning Research Intern

Bellevue, WA

Summers 2024 and 2026

- Optimized point-of-interest selection within Alitheon's visual-fingerprint object re-identification pipeline across smartphone and industrial-camera captures, cutting patch-memory footprint by 50% and improving re-identification rates by 3% over the prior state of the art.
- Built a 64k-parameter point-of-interest detector and patch embedder trained only on synthetic surfaces, paired with SAM-based masking and RANSAC/ECC alignment; raised cross-camera rank-1 identification of CPU heat spreaders from 57% to 96% (31/54 to 52/54) and reached 94% (170/180) on card identification across three cameras.
- Built a zero-shot, label-free precious-metal classification pipeline that fused OCR with neural image embeddings to detect incorrect database records, reducing manual review time by approximately 99%.
- Automated camera-station calibration for Alitheon's computer-vision-based item-authentication platform in C++, optimizing hue, saturation, and related acquisition settings on Basler and Allied Vision cameras and reducing a previously manual workflow by approximately 95%.

OpenEXA

Full-Stack Software Development Intern

Bellevue, WA

July 2023 – July 2024

- Owned the core interface for an asset-backed credit platform; documented Docker, JavaScript, and web architecture and mentored 5+ interns, improving onboarding efficiency by approximately 30%.

SELECTED PROJECTS

Multi-Agent Tanks | PyTorch, Multi-Agent RL, PPO, Attention, CUDA

- Trained one shared attention-based policy (encoder, 4-head learned team radio, GRU memory) to control every tank and base of 28-tank armies through recurrent self-play PPO, with squad, medic, and base-defense tactics emerging from reward design alone.

ChessCaptioning | Python, PyTorch, Reinforcement Learning, Attention

- Created a custom RL training pipeline and g-AWRL agent using attention-weighted mechanisms for interpretable decision-making; generated board-region heatmaps and preliminary natural-language move explanations.

CFish-Engine | C++, Search Optimization, Game AI

- Implemented alpha-beta pruning, quiescence search, move ordering, and Zobrist-hashed transposition tables; optimized bitboard representations and cache-conscious data structures for deeper search.

Self-Driving RC Rover | Embedded C/C++, FreeRTOS, ESP32-S3, ESP-NOW

- Architected a dual-controller vehicle (ESP32-S3 for sensing, safety, and wireless logic; Arduino Mega for motor control) and implemented concurrent FreeRTOS sensor and decision tasks with mutex-protected state and 25 Hz ESP-NOW remote control.

SKILLS

Machine Learning: deep learning, transformers, attention mechanisms, CNNs, reinforcement learning, representation learning, multimodal learning, zero-shot classification, OCR, computer vision, NLP, embeddings (CLIP/SigLIP), self-play / PPO

Training & Research: hyperparameter tuning, learning-rate scheduling, model training and evaluation, benchmarking, experimental design, data cleaning and curation, feature extraction, reproducible experimentation

Frameworks & Data: PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, pandas, SQL

Languages & Tools: Python, C/C++, Java, JavaScript/TypeScript, SystemVerilog, Git/GitHub, Docker, AWS, Verilator, FreeRTOS, ESP32-S3, LaTeX